

Enabling adaptable metadata thru rules-driven ontologies

Moving beyond conventional relational models and repositories to capture multiple points of view (relationships) using a “learning” model.

Examining 4 Types of Taxonomies

OVERVIEW OF EACH WITH DESIGN CONSIDERATIONS

Taxonomy Basics

- ☐ **There are four types of taxonomies**
 - Flat
 - Hierarchical
 - Faceted
 - Network

- ☐ **Some are explicit/visible (published lists), others are implicit/invisible (listings we create)**

- ☐ **There are significant design consideration when implementing each different type**

Type 1: Flat Taxonomy Structures

Domains are typically represented in a flat structure



Type 1: Flat Taxonomies

❑ Flat taxonomies group content into a controlled set of categories

- There is no inherent relationship among the categories in a flat taxonomy -- they are co-equal members of a single structure
- No interdependence, so navigation can flow within and across each category without impacting any other category

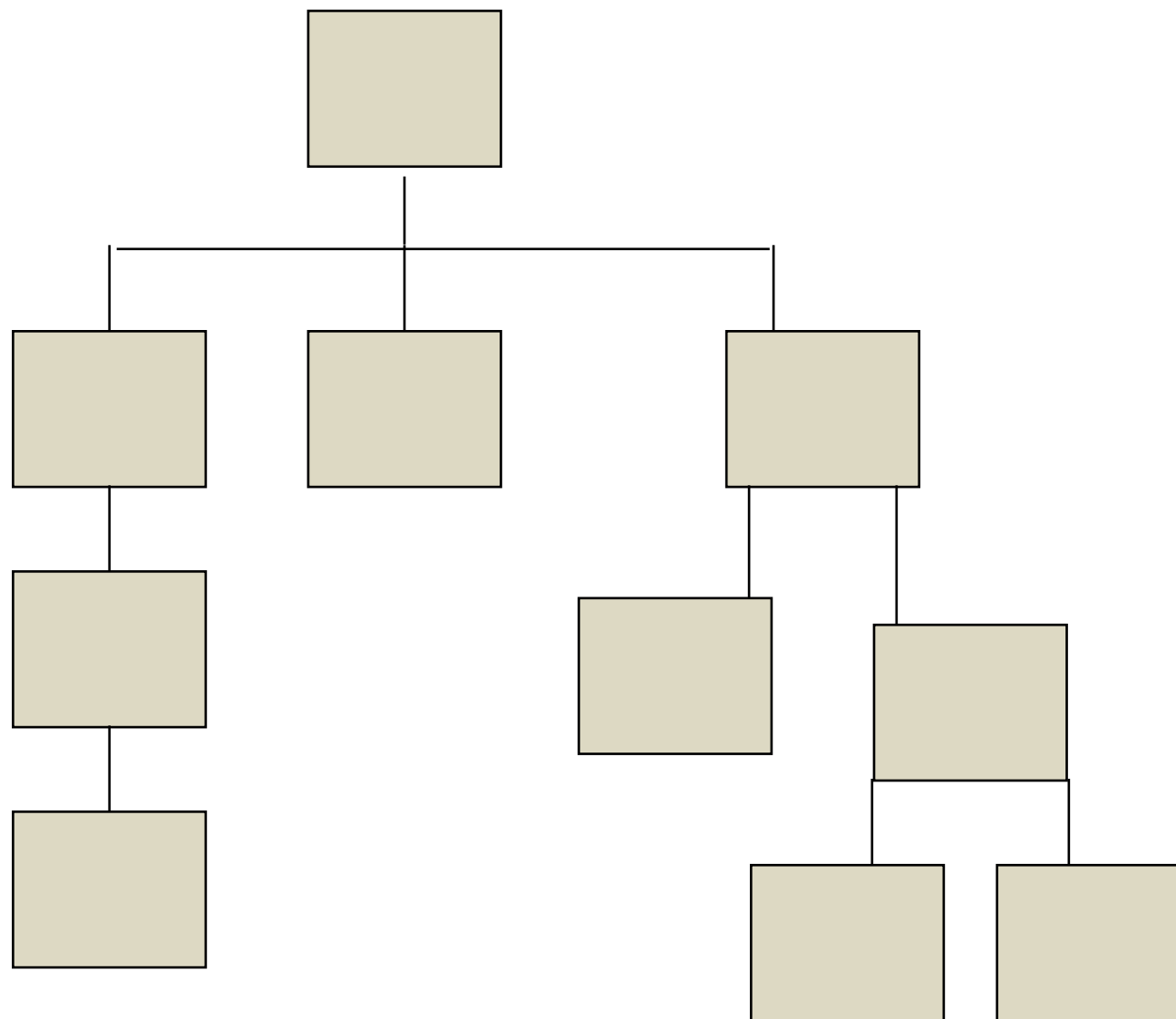
❑ Examples:

- List of domains that a business serves
- Alphabetical listings of people in a directory of expertise
- A pull-down menu of country names or geographical regions
- Simple alphabetical listings of product groupings

Type 1: Designing Flat Taxonomies

- ❑ Flat taxonomies are easy to create
- ❑ Flat taxonomies do not require complex interface design and extensive usability testing
- ❑ Design Considerations
 - Flat taxonomies used for explicit information structures generally should consist of 30 or fewer categories;
 - More than 30 categories may be presented in a flat taxonomy, if the categories are intuitive to users (i.e. lists of countries, states, languages, etc.);

Type 2: Hierarchical Taxonomy



A hierarchical taxonomy is represented as a tree data structure in a database application. The tree data structure consists of nodes and links.

In an RDBMS environment, the relationships become associations. In a hierarchical taxonomy, a node can have only one parent.

Type 2: Hierarchical Taxonomies

- ❑ **Group content into two or more levels, where hierarchical categories typically have only one parent category.**

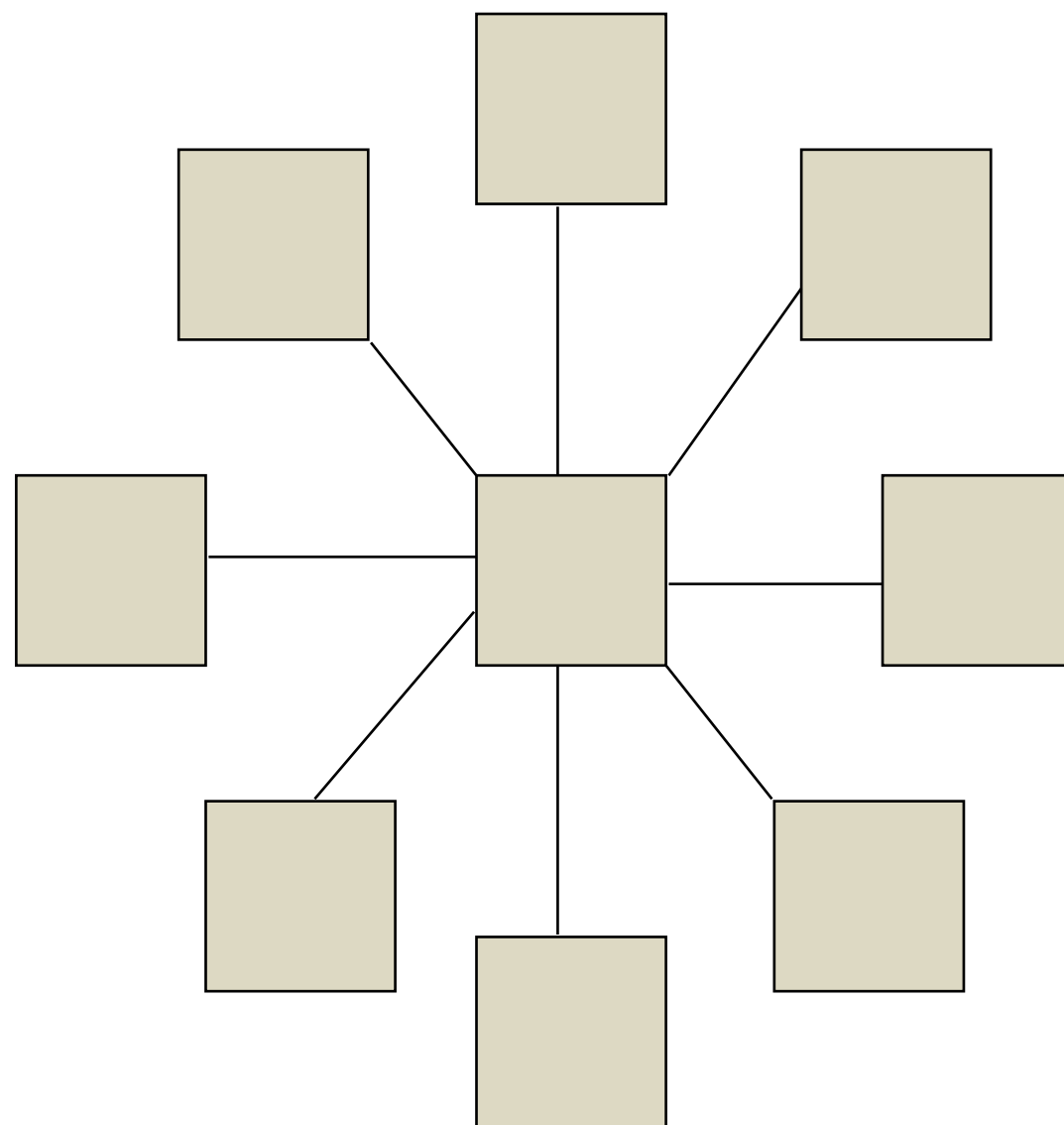
- ❑ **Relationships among categories in hierarchical taxonomies have particular meaning**
 - Relationship between a top level category & subcategory may mean group membership or refinement of the top category by a particular characteristic or feature
 - Moving up the hierarchy means expanding or broadening the category
 - Moving down the hierarchy means refining or qualifying the category

Type 2: Hierarchical Taxonomies

□ Design Considerations

- Should have content at every level -- empty categories present empty value to users
- Ideally be less than four levels deep in most cases
- Have at least two categories for each branch in the taxonomy (try to avoid branching for a single category)
- Balance breadth & depth (e.g. users must work harder to use a taxonomy three categories broad & nine deep than to use one that is seven wide and two deep)

Type 3: Facet Taxonomies



Faceted taxonomy represented as a star data structure. Each node in the star structure is linked to the center focus.

Any node can be linked to other nodes in other stars. Appears simple, but becomes complex quickly.

Type 3: Faceted Taxonomies

- ☐ Resemble flat taxonomies when implemented, but have a different structure & purpose than flat taxonomies
- ☐ There are no inherent relationships among categories in a faceted taxonomy – like a flat taxonomy
- ☐ Resemble a star structure -- all facets pertain to the center object
- ☐ All categories in a faceted taxonomy relate to a single object -
- may describe a property or a value, different views or aspects of a single topic

Type 3: Faceted Taxonomies

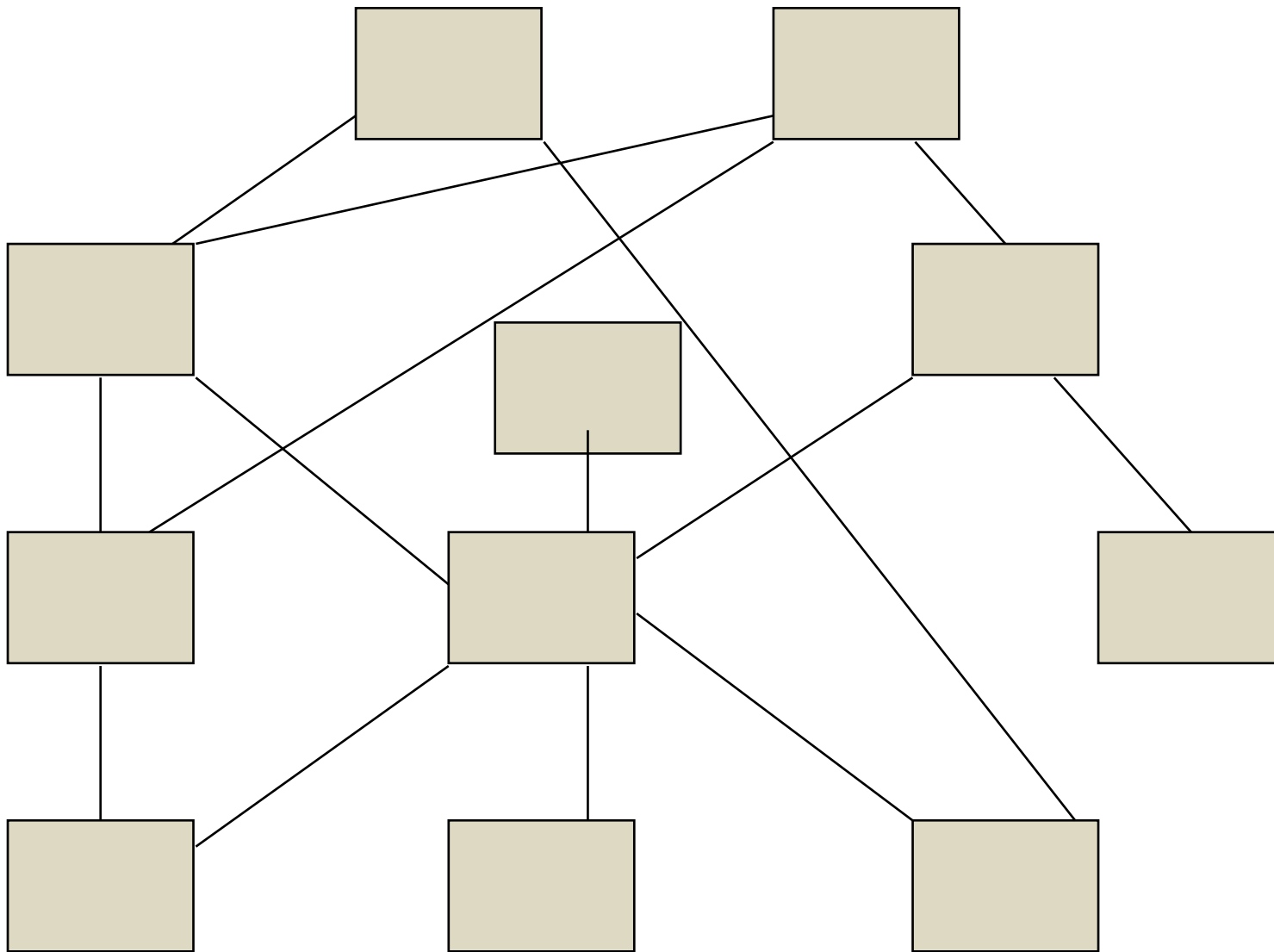
□ Design Considerations

- Characteristics of each facet should be defined fully and distinctly -- while all facets pertain to a common object, each has a distinct behavior
- Users should be able to manipulate facets distinctly -- it is important to define each facet exclusively, without overlap with other facets
- Most faceted taxonomies are implicit structures - when they are made explicit, they are generally presented as record or table formats

Designing Metadata as Faceted Taxonomy

- ❑ **The primary implicit application of faceted taxonomies today & historically is as implicit metadata records**
- ❑ **Types of taxonomies that rely on metadata today:**
 - Product & Service catalogs
 - Compliance Knowledge
 - Knowledge profiles
 - User interest profiles

Type 4: Network Taxonomies



A network taxonomy is a “plex” data structure. Each node can have more than one parent.

Any item in a “plex” structure can be linked to any other item.

In plex structures, links can be meaningful & different.

Type 4: Network Taxonomies

- ☐ **Organizes content into both hierarchical and associative categories**
- ☐ **May look like a computer network topology**
- ☐ **Many relationships among categories or “nodes”**
- ☐ **Relationships may have many different meanings**
- ☐ **Category may have more than one higher level category**
- ☐ **Any category in the taxonomy may be linked to any other category**

Type 4: Network Taxonomies

□ Design Considerations

- Different kinds of relationships may be implemented in different ways - a single approach to display all types of relationships may not be effective
- Consider leveraging rules to capture the associations rather instead of attempting complex maintenance of taxonomies
- Engage knowledge engineers to help model rules (rule sets) to simplify maintenance and ensure associations are properly managed
- Leverage Data Management tools & processes to ensure integrity of the rules sets and build these into the ongoing maintenance for rules-driven ontologies.

Type 4: Network Taxonomies

□ Examples of network taxonomies:

- Associations amongst taxonomies as broader ontologies
- Domain knowledge encapsulated as multi-faceted associations
- Personal perspectives about information and knowledge encapsulated as individual sets of associations
- **Semantic mappings of networks to infer meaning across individuals and domains**

Knowledge as a Service

FORMING A KNOWLEDGEBASE STARTING FROM AN ONTOLOGY
MODEL, AND PERFORM REASONING ON A KNOWLEDGEBASE

Goal is to create a “learning” system

Develop a system environment that provides users a way to repeatedly discover important rules or facts based on their professional judgment is absolutely an issue that research workers in the knowledge mining fields can not avoid

Association Rules Between Taxonomies

❑ Intra-dimensional rules

- Describes an association between 2-nodes within a single dimension

❑ Inter-dimensional rules

- Describes an association between nodes within a multiple dimensions

❑ Hybrid Dimensional rules

- Describes an association between nodes within a single-dimension and across multiple dimensions

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- ❑ If we can encapsulate these rules outside of the data warehouse (or knowledge repository), we can provide a significant measure of adaptability and scale for accommodating multiple perspectives (POVs).

Challenges with Conventional Data Warehousing

- ❑ **Lacks structural and/or semantic exploration of stored data**
 - Conventional DWs are “mostly” based on relational DBs, which can’t thoroughly describe associations between organizational structures & semantic data
 - Can’t prevent improper data selection (search), which leads to mining the wrong sets of patterns or rules
 - Can’t capture and manage the complexity associated with multiple POVs (For example: Product Models often lock organizations into a singular top-down model, which works well for transaction processing but creates artificial constraints for supporting variability amongst customers, suppliers, and partners).

Challenges with Conventional Data Warehousing

❑ Lacks facilities in capturing user's mining behavior

- Perception of data mining results depends on personal preference, rooted in patterned behavior (selection of parms, constraints, patterns, and semantics).
- DWs lack the ability to capture and analyze user's mining behavior to optimize future mining queries.
- The problem gets exacerbated as data volumes and/or data complexity increases.
- These behavioral patterns are effectively associations between personal preference and actual fact data and dimensional data, which are themselves sets of rules.

Challenges with Conventional Data Warehousing

❑ Lacks effective and active functions for maintaining discovered knowledge

- Conventional Data Mining is mostly passive (only activated by users queries)
- Which means that “users” have to keep track of changes in internal or external factors used in forming the queries
- Active Data Mining involves updating changes to the DW where new patterns are captured (associations created by users or discovered by the system). These associations may be new perspectives (captured from existing or new users) during query activities where repeatable patterns emerge –OR- built in by knowledge engineers to capture industry (domain) specific associations (knowledge if you will)

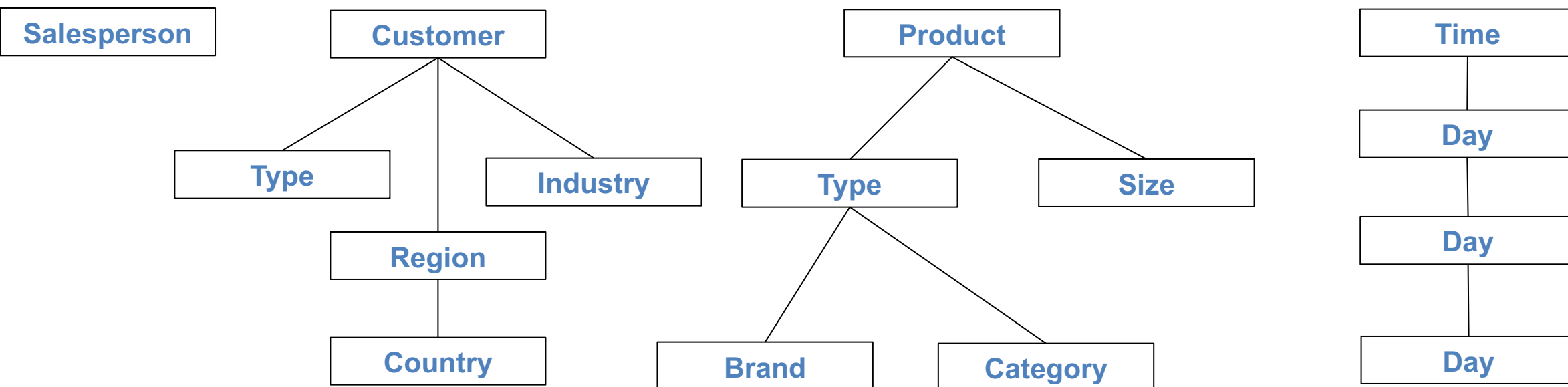
Notion of a “Learning System”

- ❑ Think of the associations as initial patterns of discovery (rules), which may be repeated (over time = reinforced) and become “permanent”.
- ❑ The advantage of using rules is that we can capture and measure the “strength” of a given pattern using probability (of occurrence).
- “think short term & long-term memory patterns instantiated as rules”

Example 1:

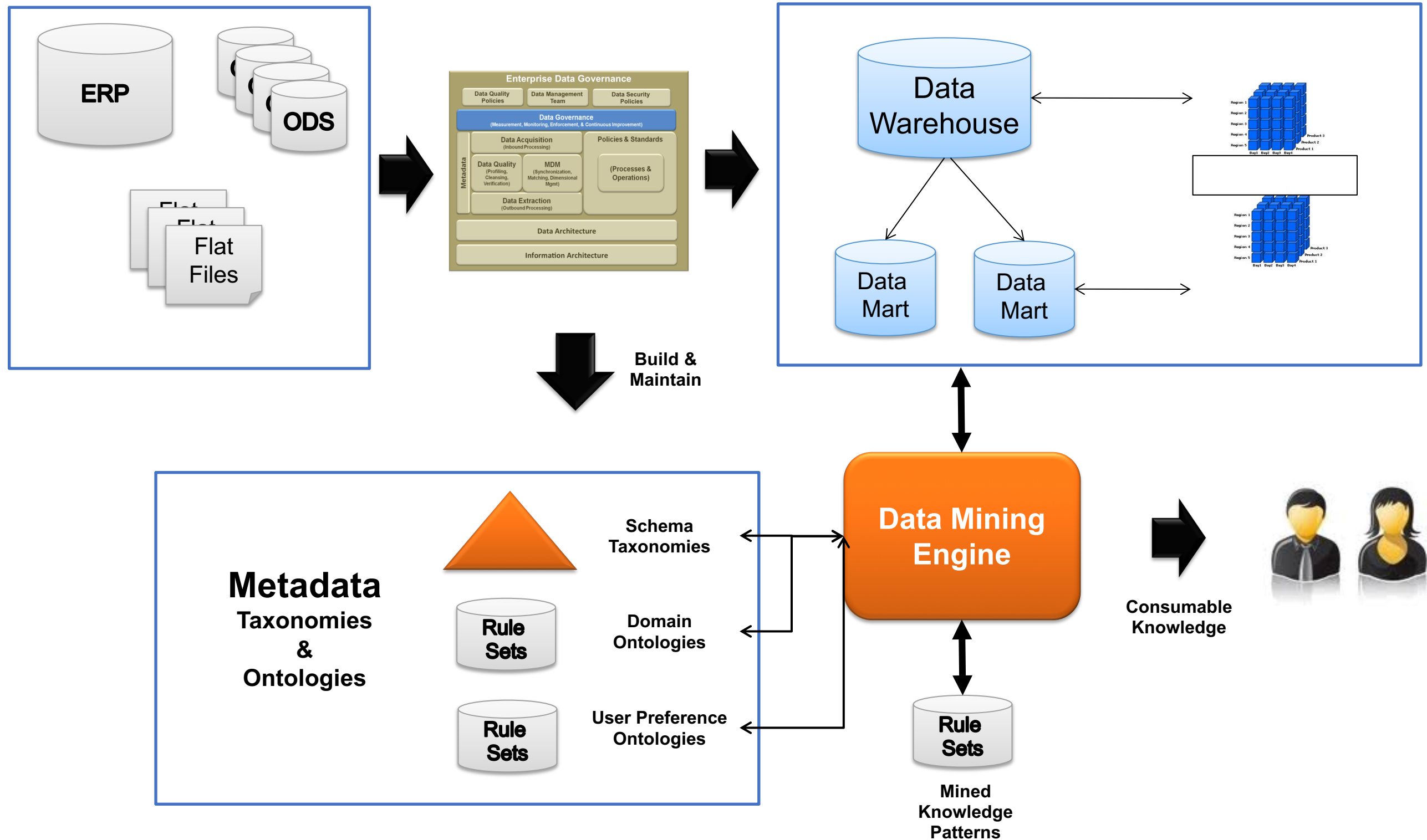
DESIGNING A MODEL TO SUPPORT ACTIVE MINING WITH 3
TAXONOMIES (BUILDING FLEXIBLE AND ADAPTABLE
ONTOLOGIES USING A RULES-DRIVEN APPROACH)

Start with 3 Taxonomies (part of a DW)



Dimensional attributes arranged star schema supporting sales information

Define a “modified” structure for our DW



Modified DW Explained

- ❑ The Schema Taxonomies are the dimensional hierarchies, schema structures, and physical keys used to map “stable” relationships between structures.
- ❑ The Domain Ontologies contain the rules for defining “variable” relationships w/in and w/o taxonomies specific to the knowledge domains.
- ❑ The Personal Preference Ontologies are similar to Domain Ontologies but are specific to each individual’s knowledge perspective.

Modified DW Explained

- ❑ **The Schema Taxonomies imply fixed levels of cardinality between the nodes within a hierarchy and between hierarchies (via the use of keys).**
- ❑ **Hierarchical relationships specify functional dependencies via parent-child with inheritance as the advantage of the structure.**
 - Product ID has a fixed set of attributes associated with each unique instance of product
 - Customer ID and Salesperson ID both have fixed sets of attributes
 - Depending on the level of normalization, attribution can be enhanced and expanded within record sets with varying levels of complexity and effort.

Modified DW Explained

- ❑ **The Domain Ontology is used to construct domain knowledge within a given subject area**
 - Sales Data and Time
 - Structure of the conceptual layer
 - Relationships amongst different customer types
 - Relationships between customers and product categories
 - Relationships between customers and demographic patterns
 - Relationships between customers and industries
- ❑ **Content is specific to certain domains and must be adaptable as domains change over time**
 - Classification and composition (bundling of components) represent relationships about what goes into a product and its compatibilities.

Modified DW Explained

❑ The Personal Preference Ontologies are used to collect historical query patterns:

- List of common attributes for query type
- Combos of attributes
- Grouping combinations of items
- Parameter settings (confidence levels)
- Frequency of occurrence settings (probability)
- Alignment with demographics
- Alignment with ethnographies
- Alignment with customer & industry attributes

Benefits of Rules-Driven Ontologies

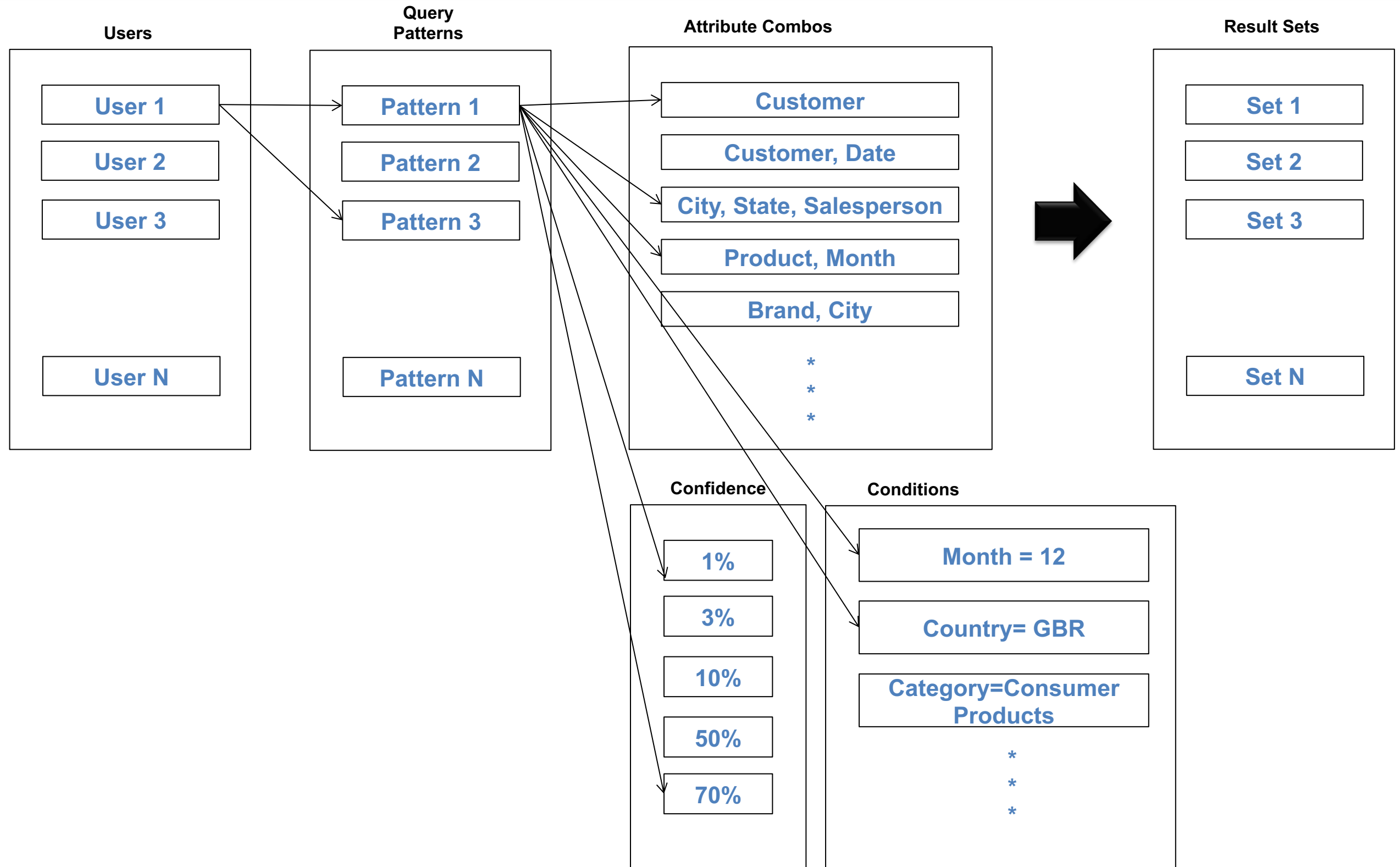
- ❑ Rules enable scaling for adaptability and individuality
- ❑ Rules can be easier to manage when addressing one or more orders of magnitudes of associations
- ❑ Overhead associated with maintaining (modifying) the hierarchies, schemas, and structures can remain relatively flat while rules scale in complexity

Example 2:

OPTIMIZING MINING QUERIES FROM A DW

CAN WE CONFIRM THAT CUSTOMERS IN A PARTICULAR INDUSTRY (CATEGORY) TEND TO PERFORM MORE RIGOROUS TESTING OF PRODUCTS ?

Step 1: Forming the Mining Query



Step 2: Building the Query

❑ Pseudo Code for the Query....

Select from multi-dimensional associations

Grouping by Time (date)

Sources (Customer, Country, Time, Month, Product Category)

From Sales (star schema)

Where Confidence = 70%, Conditions = Month+Country+Category

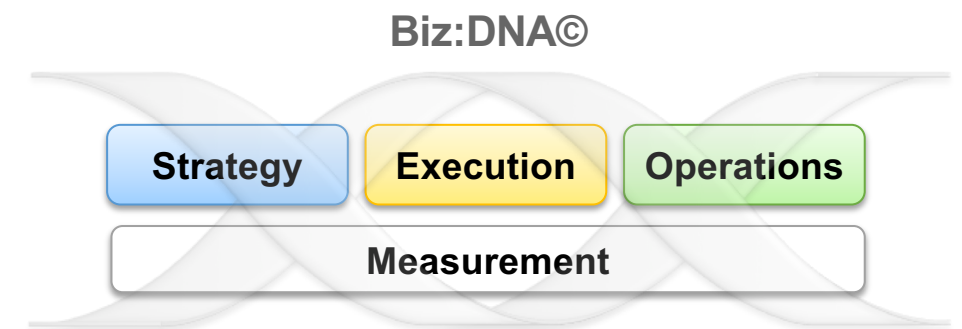
- ❑ The Associations of attributes (w/in & w/o a taxonomy) and the additional rule sets associated with Confidence & Conditions are all part of [Pattern 1](#), which was previously associated with the individual's preferences.

Step 3: Reviewing the Results

- ❑ The results should be much closer to helping the user get their job done since right knowledge was delivered in the right amount to simplify consumption and to enable understanding.
- ❑ Over time, as results sets and patterns are analyzed, additional levels of patterns may emerge revealing more opportunities to deliver value to sets of individuals (groups of job executors = new concept in defining markets). [Drives growth & expansion for the solution]

Thank You

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